TABLE S2A

CLUSTERING CONFIGURATIONS AND HYPERPARAMETERS

Field	Leiden	FlowSOM	SpatialSort	PIXIE
Inputs	47 markers	47 markers	47 markers	48 image channels
Transform	arcsinh(x/5)	arcsinh(x/5)	arcsinh(x/5)	Pixel and cell composition pipeline
Scaling	StandardScaler within each region	RobustScaler on all B004 cells	RobustScaler on all B004 cells	Channel and composition normalization
Configured and occupied clusters	300 and 300	300 requested and 299 occupied	300 requested. 229 occupied in the primary chain and 231 in the sensitivity chain	300 and 300
Implementation	K320 local Leiden with 20 within-parent merges	32×32 self-organizing map followed by Ward clustering	24 neighbors and precision 0.65	10×10 pixel map and 24×24 cell map
Sigma and learning rate	See Leiden procedure in Section 2.7	1.0 and 0.3	Not applicable	Pixel 5.0 and 0.05. Cell 2.0 and 0.3
Iterations	See Leiden procedure in Section 2.7	3 whole-cohort passes, giving 660,246 updates	100 actual sweeps per chain with dmh_iter=5	One pixel pass and 5,000 cell iterations
Seed	42	42	Sampling seed 42 primary and 43 sensitivity	42